

General Geoscience Academic Report

Spec-driven Touying setup with PKU-aware identity

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Agenda

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PART

01

Background

Frame the geoscience problem, why it matters, and what evidence the talk will use.

Study Area & Geologic Setting

A good geoscience report first anchors the audience in place, scale, geomorphic context, and available observations.

- Show the geographic setting and relevant tectonic, climatic, or geomorphic controls.
- Mark observation footprints, field sites, profiles, or regions of interest.
- Use captions to state what the audience should read from the map.

Placeholder for a study-area map, DEM, satellite image, or geologic setting figure.

Data & Observations

Data pages should make the evidence base explicit before interpretation begins.

Data sources

- Remote-sensing imagery, DEMs, geologic maps, field measurements, or sensor time series.
- Acquisition period, spatial resolution, uncertainty, and preprocessing status.
- External datasets and citations required for reproducibility.

Use in this report

- Separate raw observations from derived products.
- Use one page for overview, then figure-first result pages for details.

PART

02

Method

Describe the processing chain or model at the level needed for the audience to trust the results.

Method Workflow

Method slides should expose the logical chain from observations to geoscience interpretation.

- Define inputs and preprocessing choices.
- State the core model, assumptions, or classification logic.
- Show validation, sensitivity checks, and uncertainty propagation.



Placeholder for workflow diagram, model schematic, or inversion pipeline.

PART

03

Results

Use figure-first pages: one claim, one primary visual, and a small number of interpretation points.

Key Results

The result page API is intentionally generic so it can host maps, profiles, time series, model outputs, or comparison plots.

- Lead with the claim, not the plotting process.
- Tie visual patterns back to physical interpretation.
- Call out uncertainty or alternative explanations where they affect the conclusion.



Placeholder for main geoscience result figure.

Uncertainty & Limitations

A robust academic report separates interpretation from uncertainty and next validation steps.

Assumptions

- The placeholder workflow assumes datasets are co-registered and comparable.
- The chosen figure layout assumes one dominant message per result page.

Limitations

- Venue-specific styling is represented as setup metadata, but the low-level theme palette is not yet fully tokenized.
- Real reports still need project-specific data citations and figure assets.

Next checks

- Replace placeholders with real maps and result figures.
- Add venue-specific examples for AGU, EGU, and routine group meetings.

Conclusions

This first scaffold proves the workflow: spec, preset setup, semantic component, example, and compile check.

- The deck now uses a public `lib/geo-presentation.typ` API.
- Language, venue, objective, and density are represented by a setup object.
- Geoscience-specific pages are semantic wrappers around the existing Touying backend.
- Future styling work can refine the backend without changing user-facing examples.

References

- [1] Touying package documentation. Typst Universe. Used as the presentation framework backend.
- [2] Awesome-Marp project by favourhong. Used as styling inspiration for reusable layouts and theme-color switching.
- [3] Metropolis Beamer theme. Used as academic-presentation inspiration for restrained sectioning and navigation.
- [4] Project specs under `specs/`. Used as the source-of-truth contract for future AI-agent work.

Thank You

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